Agenda.

→ Vector Embedding

→ Similarity.

⇒ Embedding

↳ Way of representing data (text / images / audio or video) in a multi-dimensional space, where semantically similar items are placed together. (Similar meaning)

Dog Puppy

Embeddings are needed for:

→ Semantic search

→ Recommendation systems.

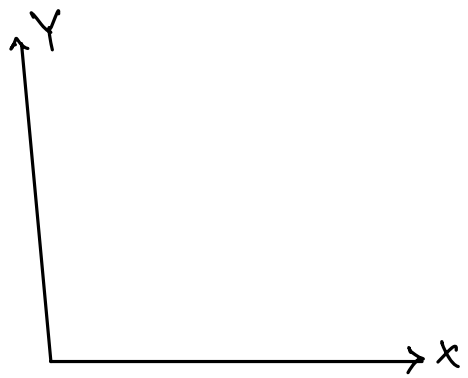
→ Clustering

Dog → 60

Cat → 50

Computer → 02

Laptop → 04



Vector

↳ list of numbers

[10]

[1, 2]

[1 2 3]

[- - - - - -]

dog → [- - - - - - - - - - - -]

↳ 1576

Vector Embedding

Numerical representation of data that captures its meaning & context.

Text Embedding

Converts text | word | sentence | paragraph | document into vector of numbers.

dog → [0.7 0.45 0.9]

puppy → [0.65 0.48 0.91]

computer → [0.07 0.99 0.15]

laptop → [0.05 0.97 0.16]

Embeddings are designed such that:

Similar meaning → Close vectors.

Different meaning → far apart vectors.

Sparse (vs) Dense Embedding

Vectors where most values are ZERO.

Very less no. of zeroes.

[- - - - - - 0 0 0 0 0 0 0 0 0 0 0 - - -]

In Embedding space, Vectors present closely to each other will have similar meanings.

Input : "Python is widely used in Machine learning"

↓ Vector Embedding

STEPS

1. Tokenization
   - Text is broken down into smaller pieces called as tokens.

2. Token Representations
   - Each token is mapped into their vector representations.

3. Contextual Processing
   - Model processes the sentence & learns contextual meaning.

bank → River Bank
bank → Financial Bank.

④ final Embedding
→ Vector embedding.

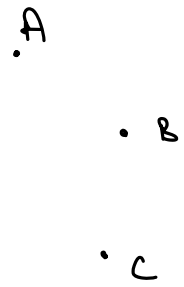
gpt-embedding
→ Small
→ large

Small (vs) large Embedding models.

Small → less dimensions → fast, Cheaper, less memory → Low Quality

large → High dimensions → Slow, Costly, High Memory → High Quality.

Similarity Search.

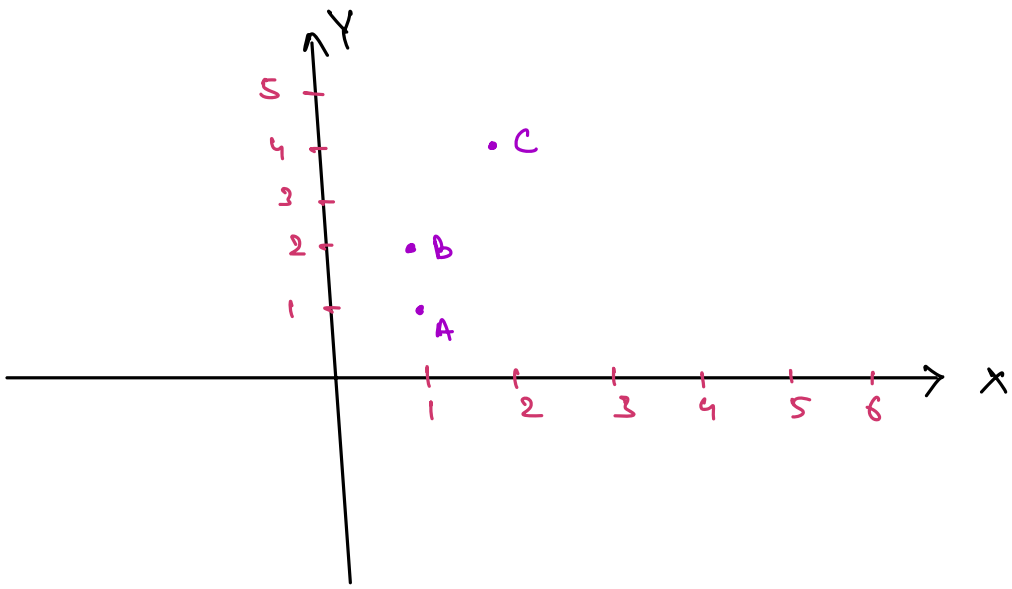


x y

A : [1, 1]

B : [1, 2]

C : [2, 4]



Euclidean Distance :

$$\left.\begin{array}{l} A(x_1, y_1) \\ B(x_2, y_2) \end{array}\right\} \Rightarrow d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$d(A,B) = \sqrt{(1-1)^2 + (1-2)^2}$$

$$= \sqrt{0 + 1^2}$$

$$= 1$$

$$d(A,C) = \sqrt{(1-2)^2 + (1-4)^2}$$

$$= \sqrt{1^2 + 3^2}$$

$$= \sqrt{10} = 3.3\ldots$$

Distance $\uparrow$ $\Rightarrow$ Similarity $\downarrow$

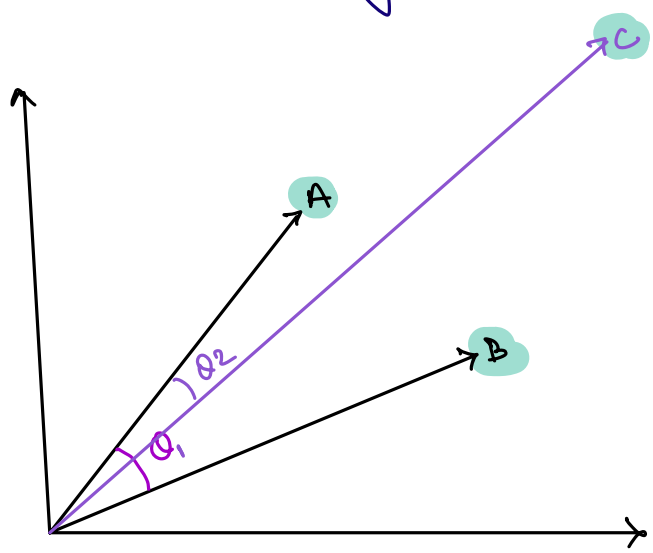
Cosine Similarity.

Measures the angle b/w 2 vectors.

Sin θ

Cos θ

tan θ

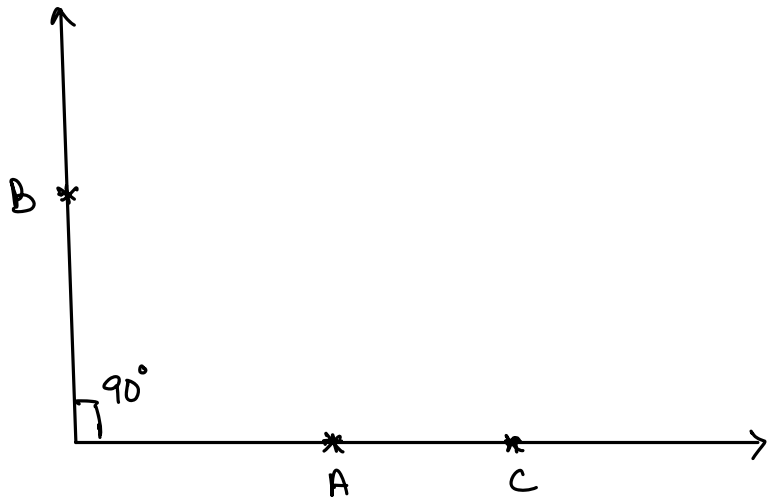


Cosine similarity focuses more on the direction than the exact magnitude.

$\cos 0 = 1$

$\cos 90 = 0$

$\cos 180 = -1$



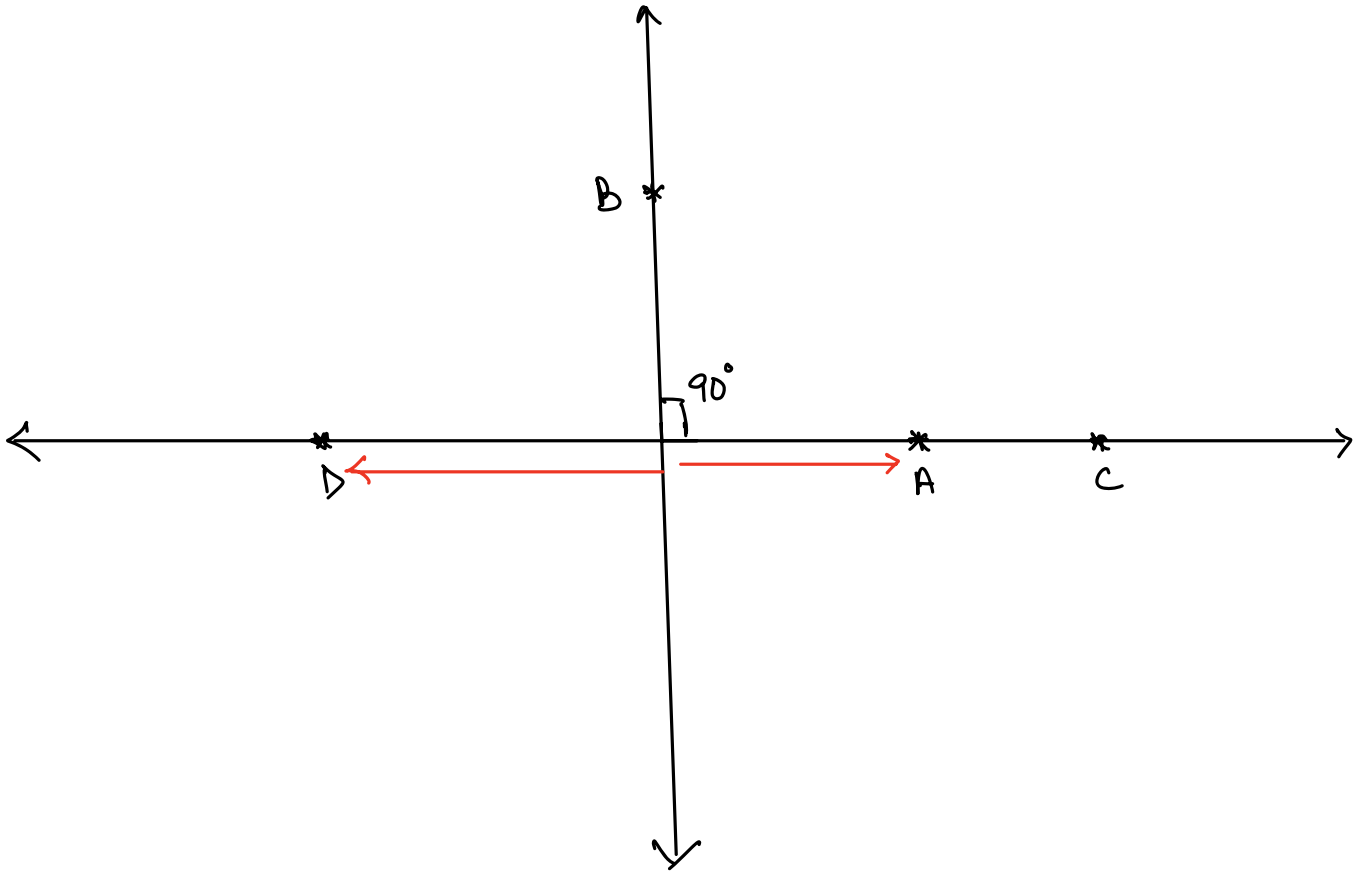
$$\text{Cosine distance} = 1 - \cos\theta$$

$$\text{Cosine distance}(A, B) = 1 - \cos 90$$

$$= 1$$

$$\text{Cosine distance}(A, C) = 1 - \cos 0$$

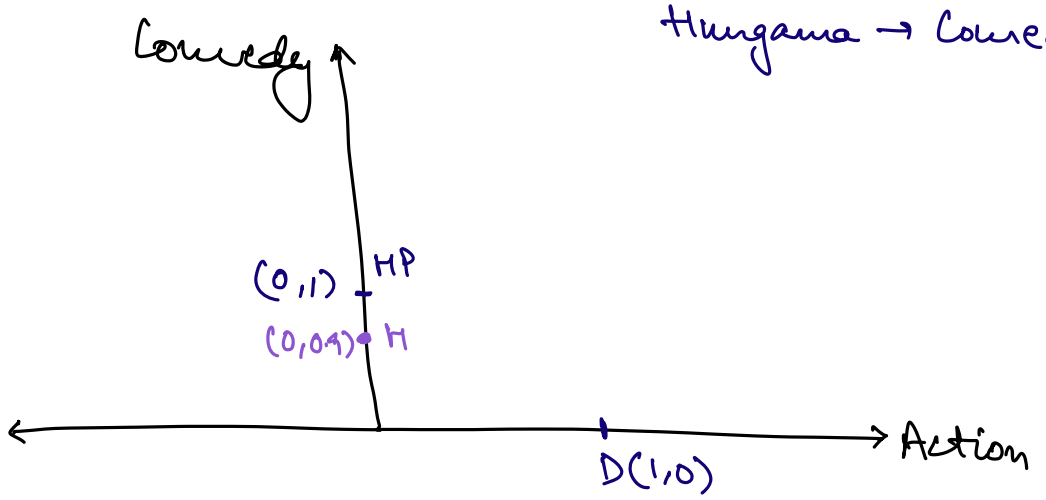
$$= 0$$



$$\text{Cosine distance}(A, D) = 1 - \cos 180$$
$$= 1 - (-1)$$
$$= 2$$

Movie → Action
Movie → Comedy

Dhurandar → Action
Hera Pheri → Comedy
Hungama → Comedy



Create Embedding

External API (OpenAI)

Local | Open Models